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The Efficiency Cascade: From Starlight to Steak

Four numbers, four kinds of loss: what physics, plants, food webs, and cud have in common

By Lee McCulloch-James

EVERY energy transformation in nature throws most of the energy away. A fusing star, a growing leaf, a food chain, and a cow's stomach, and lining them up side by side shows that their"efficiency" is not one idea but at least four different ones wearing the same name.

Fusion: The Nuclear Floor

At the Sun's core, four hydrogen nuclei fuse into one helium-4 nucleus. The helium nucleus weighs about 0.7% less than the four protons that made it, and that missing mass is released as energy, $E = mc^2$. This is fixed by nuclear binding energy: the same 0.7% everywhere hydrogen fuses to helium, with no room for variation.

But a star does not convert 0.7% of itself into starlight. Only its hot, dense core, roughly the inner tenth of the Sun's mass, ever fuses; the outer nine-tenths sits unburned for the star's entire life. Multiplying the two factors gives the lifetime-integrated efficiency: about 6.8×10^{-4}, under a tenth of one percent, of the Sun's rest-mass energy will ever become starlight.

Photosynthesis: Catching the Light

A leaf faces the opposite problem. Of all the sunlight striking it, 98% to 99% is reflected or simply absorbed as heat. Chlorophyll only captures specific wavelengths, mainly red and blue, missing most of the green, and every step of the biochemistry that follows, electron transport, then the Calvin cycle, loses more energy again before any of it is locked into a sugar bond.

Real-world plants convert only about 1% to 2% of incident sunlight into stored chemical energy. The theoretical ceiling, assuming perfect biochemistry, is around 4.6% for ordinary (C3) plants and 6% for the more efficient (C4) plants such as maize and sugarcane (Zhu, Long, and Ort, 2008).

BY THE NUMBERS

Fusion, single event (H→He)	0.7%
Fusion, whole star, lifetime	6.8×10^{-4}
Photosynthesis, realized	1–2%
Photosynthesis, theoretical max	4.6–6%
Trophic transfer (Lindeman's rule)	$\sim 10\%$
Rumination, methane loss alone	4–8% of GE
Rumination, gross-to-net (typical)	$\sim$ 25–40%

Trophic Transfer: The Ten Percent Rule

Moving up a food chain is a different kind of accounting entirely. Raymond Lindeman's 1942 rule of thumb holds that only about 10% of the energy stored in one trophic level's total biomass ends up incorporated into the next level's biomass, with the figure ranging roughly 5% to 20% across real ecosystems (Lindeman, 1942).

The rest is lost in multiple bottlenecks; spent on respiration, movement, and body heat, or walks away as prey never caught, killed but not eaten, or eaten but not digested. Unlike the fusion number, this is a statistical average over enormous numbers of separate, sloppy transactions.

Rumination: Beating the Average

Cellulose, the main structural material of plants, cannot be digested by any mammalian enzyme. A cow's answer is the rumen: a fermentation chamber where microbes break cellulose down before the cow re-chews and re-swallows the cud. Tracked through the standard bioenergetics cascade, gross energy in the forage first loses the largest single share to feces, since fermentation is never complete. Methane production, a direct by-product of microbial fermentation, then carries away a further 4% to 8% of the original gross energy, and combined gas and urinary losses together average about 18% of the digestible energy that remains (Oregon State University, *Principles of Animal Nutrition*, ch. 17). Heat generated by fermentation itself, the heat increment, reduces the total once more before what is left, net energy, becomes available for growth, maintenance, or milk.

The overall gross-to-net efficiency for a forage-fed animal is at 25% to 40%, well above the ecosystem-level ten percent rule. Rumination is a biological adaptation for beating that average, purchased at the cost of the methane emissions.

Same Word, Different Physics

Four numbers, four different kinds of constraint. Fusion is fixed by nuclear binding energy, unimprovable by evolution or engineering. Photosynthesis is an optical and biochemical ceiling, set by which wavelengths chlorophyll absorbs and how much the reaction machinery afterward wastes. The trophic figure is an emergent statistical average over millions of independent predator-and-prey transactions, not a property of any one organism. Rumination is a targeted physiological fix, evolution's answer to a digestive problem, that outperforms the ecosystem-wide average it sits inside. Calling all four "efficiency" is convenient shorthand; only one of them, the 0.7% locked into every hydrogen nucleus that has ever fused, could not in principle be otherwise.

FURTHER READING

Lindeman, R. L. (1942). The Trophic-Dynamic Aspect of Ecology. *Ecology* 23(4), 399–418.

Zhu, X.-G., Long, S. P., and Ort, D. R. (2008). What Is the Maximum Efficiency With Which Photosynthesis Can Convert Solar Energy Into Biomass? *Curr. Opin. Biotechnol.* 19(2), 153–159.

Britannica. Biosphere: Efficiency of Solar Energy Utilization.

Oregon State University. *Principles of Animal Nutrition*, Chapter 17: Bioenergetics. Open Educational Resource.

KEY TERMS

Nuclear fusion. Combination of light nuclei into heavier ones, releasing energy from the resulting loss in mass.
Trophic level / ecological efficiency. A position in a food chain, counted by energy transfers from the original solar input; the fraction passed to the next level.
Rumination. Re-chewing of fermented cud from the rumen, raising energy extraction from fibrous plant material.
Gross, digestible, metabolizable, net energy. An animal's energy accounting, each stage found by subtracting one loss: feces, then gas and urine, then heat.